

Bret Alexander Beheim

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Research areas: behavioral ecology; cultural evolution; social learning and cultural transmission; human behavioral ecology; quantitative and computational methods; causal inference.

Academic Appointments

Group Leader, MOSAIC Research Group Max Planck Institute for Evolutionary Anthropology, Leipzig, Germany	2020–present
Senior Researcher, Department of Human Behavior, Ecology, and Culture Max Planck Institute for Evolutionary Anthropology, Leipzig, Germany	2016–2020
Postdoctoral Researcher, Department of Anthropology University of New Mexico, USA	2012–2016

Education

Ph.D. in Ecology University of California, Davis, USA	2007–2012
M.A. in Social Sciences University of Chicago, Illinois, USA	2005–2006
B.A. in History and Anthropology, with distinction in the major University of California, Santa Barbara, USA	2000–2004

Research Funding and Awards

- **2016–present.** MPI-EVA Senior Researcher Funding Line: EUR 50,000 startup plus EUR 50,000 annual budget over five years; extended by seven years in 2020.
- **2015.** Co-Investigator, “The Human Life Course and the Biodemography of Aging,” National Institutes of Health / National Institute on Aging, \$590,659.
- **2014.** Co-Investigator, administrative supplement to “The Human Life Course and the Biodemography of Aging,” NIH / NIA, \$15,000.
- **2014.** Co-Investigator, “Surviving the Flood: Vulnerability, Risk Management, and Resilience among Tsimane Forager-Farmers of Bolivia,” National Science Foundation, \$126,000.
- **2009.** Graduate Fellowship Block Grant, University of California, \$10,800.
- **2007.** Graduate Scholars Fellowship in Ecology, University of California, \$34,022.

Book

Beheim, B.A. & Smaldino, P.E. *Mathematical Foundations of Cultural Evolution*. Princeton University Press. Under contract; approximately 400 pages; manuscript due December 2026.

Complete Publication List

1. Sánchez-Amaro, A., Ebel van Wijk, S.J., Molenaar, C. et al. (2026) EVApeCognition: An 18-Year Dataset of Great Ape Cognition *Nature Scientific Data* 13(805). DOI: 10.1038/s41597-026-07191-6
2. Inchauste, L., de Lamballerie, X., Priet, S., et al. (2026) Exceptionally low mortality despite widespread COVID-19 infection among Indigenous Tsimane and Mosen of Bolivia *Social Science & Medicine* 404(119493). DOI: 10.1016/j.socscimed.2026.119493
3. Sobchuk, O., Beheim, B. (2025) Does literature evolve one funeral at a time? *Proceedings of the Royal Society, B* 292(2040), 20242033. DOI: 10.1098/rspb.2024.2033

4. Martin, J.S., Beheim, B., Gurven, M., Kaplan, H., Stieglitz, J., Trumble, B.C., Hooper, P., Cummings, D., Eid Rodriguez, D., Jaeggi, A.V. (2025) Indirect genetic effects among neighbors promote cooperation and accelerate adaptation in a small-scale human society *Science Advances* 11,eads3129. DOI: 10.1126/sciadv.ads3129
5. Daw, F., Beheim, B.A., Wascher, C.A.F. (2025) Evolutionary drivers of caching behaviour in corvids *Animal Cognition* 28(17). DOI: 10.1007/s10071-025-01938-1
6. Wascher, C.A.F., Waterhouse, G., Beheim, B.A. (2025) Vocal mimicry in corvids *Animal Cognition* 28(88). DOI: 10.1007/s10071-025-02006-4
7. Beheim, B. (2025) Opening strategies in the Game of Go from feudalism to superhuman AI *Evolutionary Human Sciences* 7(e28). DOI: 10.1017/ehs.2025.10016
8. Pérez Velilla, A., Beheim, B., Smaldino, P.E. (2025) The development of risk behaviors and their cultural transmission. *Psychological Review* Advance online publication. DOI: 10.1037/rev0000599
9. Arnoff, J.E., Jenkins, C.L., Garcia, A.R., Koebele, S.V., Ghafoor, S., Wollard, K.L., Charifson, M., Suarez, I.M., Eid Rodriguez, D.E., Beheim, B., *et al.* (2025) Inflammaging is minimal among forager-horticulturalists in the Bolivian Amazon *Proceedings of the Royal Society, B* 292(2053). DOI: 10.1098/rspb.2025.1111
10. Anderson, A.S., Blackwell, A., Sutherland, M.L., Kraft, T., Sutherland, J., Beheim, B., *et al.* (2025) Childhood skeletal lesions common in prehistory are present in living forager-farmers and predict adult markers of immune function *Science Advances* 11(29). DOI: 10.1126/sciadv.adw3697
11. Getz, M.J., Aronoff, J.E., Jenkins, C.L., Ghafoor, S., Vazquez, J., Appel, N.T., Gatz, M., Cummings, D.K., Hooper, P.L., Beheim, B., *et al.* (2025) Higher blood lipid levels after the transition to menopause in two forager-horticulturalist populations *Evolution, Medicine, and Public Health* 13(1), pp.201-214. DOI: 10.1093/emph/eoaf020
12. Colby, A.E., Jud, D.C., Baettig, V., Martin, J.S., Scaff, C., Gurven, M.D., Trumble, B.C., Beheim, B.A., Hooper, P.L., Cummings, D.K., Kaplan, H., Stieglitz, J., Cary Ista, A., Jaeggi, A.V. (2025) Oxytocin varies across the life course in a sex-specific way in a human subsistence population *Proceedings of the National Academy of Sciences* 122(51), pp. e2509977122. DOI: 10.1073/pnas.2509977122
13. Beheim, B.A., Bell, A.V. (2024) Why cultural distance can promote — or impede — group-beneficial outcomes *Evolutionary Human Sciences*, 6(e14). DOI: 10.1017/ehs.2024.8
14. Dalzero, A., Beheim, B.A., Ross, C., Hooper, P., Kaplan, H., Gurven, M., Lukas, D. (2024) Cross-cousin marriage among Tsimane forager-horticulturalists during demographic transition and market integration *Evolutionary Human Sciences*, 6(e18). DOI: 10.1017/ehs.2024.11
15. Ready, E., Ross, C.T., Beheim, B., Parrott, J. (2024) Indigenous food production in a carbon economy *Proceedings of the National Academy of Sciences*, 121(32), e2317686121. DOI: 10.1073/pnas.2317686121
16. Trumble, B.C., Schwartz, M., Ozga, A., Schwartz, G.T., Stojanowski, C.M., Jenkins, C.L., Kraft, T.S., *et al.* (2024) Poor Oral Health Is Associated With Inflammation, Aortic Valve Calcification, and Brain Volume Among Forager-Farmers *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences*, 79(5), pp.1-8. DOI: 10.1093/gerona/glae013
17. Kaplan, H.K., *et al.* (2023) Brain volume, energy balance and cardiovascular health in two non-industrial South American populations *Proceedings of the National Academy of Sciences*, 120(13), e2205448120. DOI: 10.1073/pnas.2205448120
18. Ross, C.T., Hooper, P., Smith, J.E., Jaeggi, A.V., Smith, E.A., Gavrillets, S., *et al.* (2023) Reproductive inequality in humans and other mammals. *Proceedings of the National Academy of Sciences*, 120(22), e2220124120. DOI: 10.1073/pnas.2220124120
19. Mattison, S.M., Liu, R., Mattison, P.M., Beheim, B.A., Shenk, M.K., Blumenfield, T., Sum, C.Y., Seabright, E., Alami, S. (2023) Gender disparities in material and educational resources differ by kinship system, *Proceedings of the Royal Society B*, 378(1883), 20220299. DOI: 10.1098/rstb.2022.0299
20. Kraft, T.S., *et al.* (2023) Metapopulation dynamics of infectious disease (SARS-CoV-2) transmission in a small-scale human society *PLOS Biology*, 21(8), e3002108. DOI: 10.1371/journal.pbio.3002108
21. Trumble, B., Pontzer, H., Stieglitz, J., Cummings, D.K., Wood, B., Emery Thompson, M., Raichlen, D., Beheim, B., Yetish, G., Kaplan, H., Gurven, M. (2023) Energetic costs of testosterone in two physically active foraging populations. *American Journal of Human Biology*, e23949. DOI: 10.1002/ajhb.23949

22. Stieglitz, J., Buoro, Y., Beheim, B., Trumble, B., Kaplan, H., Gurven, M. (2023) Labour's pain: strenuous subsistence work, mechanical wear-and-tear and musculoskeletal pain in a non-industrialized population *Proceedings of the Royal Society B*, 290(1998), 20222497. DOI: 10.1098/rspb.2022.2497
23. Hillemann, F., Beheim, B.A., Ready, E. (2023) Socio-economic predictors of Inuit foraging strategies and their implications for climate change adaptation *Philosophical Transactions of the Royal Society B*, 378(1889), 20220395. DOI: 10.1098/rstb.2022.0395
24. Trumble, B.C., *et al.* (2023) Apolipoprotein-ε4 is associated with higher fecundity in a natural fertility population *Science Advances*, 9(32), eade9797. DOI: 10.1126/sciadv.ade9797
25. Fedorova, N., McElreath, R., Beheim, B. (2022) The complex life course of mobility: Quantitative description of 300,000 residential moves in 1850-1950 Netherlands. *Evolutionary Human Sciences*, 4, E39. DOI: 10.1017/ehs.2022.33
26. Kraft, T.S., Cummings, D., Venkataraman, V., Alami, S., Beheim, B., Hooper, P., Seabright, E., Trumble, B., Stieglitz, J., Kaplan, H., Endicott, K.L., Endicott, K.M., Gurven, M. (2022) Female cooperative labor networks in hunter-gatherers and horticulturalists. *Philosophical Transactions of the Royal Society B*, in press. DOI: 10.1098/rstb.2021.0431
27. Minocher, R., Atmaca, S., Bavero, C., McElreath, R., Beheim, B. (2021) Estimating the reproducibility of social learning research published between 1955 and 2018. *Royal Society Open Science*, 8:210450. DOI: 10.1098/rsos.210450
28. Garcia, A.R., *et al.* (2021) APOE4 is associated with elevated blood lipids and lower levels of innate immune biomarkers in a tropical Amerindian subsistence population. *eLife*, 10:e68231. DOI: 10.7554/eLife.68231
29. Tran, N.H., Kucharský, S., Waring, T., Atmaca, S., Beheim, B. (2021) Limited scope for group coordination for stylistic variations in kolam art. *Frontiers in Psychology* 12, p.4770. DOI: 10.3389/fpsyg.2021.742577
30. Beheim, B., *et al.* (2021) Treatment of missing data determines conclusions regarding moralizing gods. *Nature*, 595. DOI: 10.1038/s41586-021-03655-4
31. Jaeggi, A.V., *et al.* (2021) Relative wealth and inequality associate with health in a small-scale subsistence society. *eLife*, 10:e59437. DOI: 10.7554/eLife.59437
32. Rowan, C., *et al.* (2021) Very Low Prevalence and Incidence of Atrial Fibrillation among Bolivian Forager-Farmers. *Annals of Global Health*, 87(1), p.18. DOI: 10.5334/aogh.3252
33. Tran, N.H., Waring, T., Atmaca, S., & Beheim, B. (2021) Entropy trade-offs in artistic design: A case study of Tamil kolam. *Evolutionary Human Sciences*, 3(E23). DOI: 10.1017/ehs.2021.14
34. Broesch, T., *et al.* (2020) Navigating cross-cultural research: methodological and ethical considerations. *Proc. R. Soc. B*. 287:20201245. DOI: 10.1098/rspb.2020.1245
35. Wood, *et al.* (2020) Step Counts From Satellites: Methods for Integrating Accelerometer and GPS Data for More Accurate Measures of Pedestrian Travel. *Journal for the Measurement of Physical Behaviour*. 3(1), pp. 58-66. DOI: 10.1123/jmpb.2019-0016
36. Borgerhoff Mulder, *et al.* (2019) Differences between sons and daughters in the intergenerational transmission of wealth. *Philosophical Transactions of the Royal Society B*. 374:20180076. DOI: 10.1098/rstb.2018.0076
37. Ross, C. T., *et al.* (2018) Greater wealth inequality, less polygyny: Rethinking the polygyny threshold model. *Journal of The Royal Society Interface*. 15(144). DOI:10.1098/rsif.2018.0035
38. Trumble, Benjamin C., *et al.* (2018) Parental hormones are associated with crop loss and family sickness following catastrophic flooding in lowland Bolivia. *Physiology & Behavior*. 193(A), pp. 101-107. DOI: 10.1016/j.physbeh.2018.02.028
39. Schniter, Eric, *et al.* (2018) Information transmission and the oral tradition: Evidence of a late-life service niche for Tsimane Amerindians. *Evolution and Human Behavior*. 39.1: 94-105. DOI: 10.1016/j.evolhumbehav.2017.10.006
40. Kaplan, H., Thompson, R.C., Trumble, B.C., Wann, L.S., Allam, A.H., Beheim, B., Frohlich, B., Sutherland, M.L., Sutherland, J.D., Steiglitz, J. *et al.* (2017) Coronary atherosclerosis in indigenous South American Tsimane: a cross-sectional cohort study. *The Lancet*. 389(10080), pp. 1730-1739. DOI: 10.1016/S0140-6736(17)30752-3
41. Gurven, M., Fuerstenberg, E., Trumble, B., Steiglitz, J., Beheim, B., Davis, H., Kaplan, H. (2017) Cognitive performance across the life course of Bolivian forager-farmers with limited schooling. *Developmental Psychology*. 53(1), pp. 160-176. DOI: 10.1037/dev0000175

42. Blackwell, A.D., Urlacher, S., Beheim, B., von Rueden, C., Jaeggi, A., Stieglitz, J., Trumble, B.C., Gurven, M., Kaplan, H. (2017) Growth references for Tsimane forager-horticulturalists of the Bolivian Amazon. *American Journal of Physical Anthropology*. 162(3), pp. 441-461. DOI: 10.1002/ajpa.23128
43. Gurven, M., Stieglitz, J., Trumble, B., Blackwell, A.D., Beheim, B., Davis, H., Hooper, P., Kaplan, H. (2017) The Tsimane Health and Life History Project: Integrating anthropology and biomedicine. *Evolutionary Anthropology*. 26(2) pp. 54-73. DOI: 10.1002/evan.21515
44. Trumble, B. C., Stieglitz, J., Blackwell, A. D., Allayee, H., Beheim, B., Finch, C. E., Gurven, M., and Kaplan, H. *et al.* (2017) "Apolipoprotein E4 is associated with improved cognitive function in Amazonian forager-horticulturalists with a high parasite burden." *The FASEB Journal*. 31.4, pp. 1508-1515. DOI: 10.1096/fj.201601084R
45. Gurven, M., Costa, M., Eid Rodriguez, D., Stieglitz, J., Beheim, B., Trumble, B., & Kaplan, H. (2016). Health costs of reproduction are minimal despite high fertility, mortality and subsistence lifestyle. *Nature Scientific Reports*. 6(30056). DOI: 10.1038/srep30056
46. Jaeggi, A.V., Hooper, P.L., Beheim, B., Kaplan, H., Gurven, M. (2016) Reciprocal exchange patterned by market forces helps explain cooperation in a small-scale society. *Current Biology*. 26(16), pp. 2180-2187. DOI: 10.1016/j.cub.2016.06.019
47. Gurven, M., Trumble, B.C., Stieglitz, J., Yetish, G., Cummings, D., Blackwell, A.D., Beheim, B., Kaplan, H., Pontzer, H. (2016) High resting metabolic rate among Amazonian forager-horticulturalists experiencing high pathogen burden. *American Journal of Physical Anthropology*. 161(3), pp. 414-425. DOI: 10.1002/ajpa.23040
48. Blackwell, A., Trumble, B.C., Maldonado Suarez, I., Stieglitz, J., Beheim, B., Snodgrass, J., Kaplan, H., Gurven, M. (2016) Immune Function in Amazonian Horticulturalists. *Annals of Human Biology*. 43(4), pp. 382-396. DOI: 10.1080/03014460.2016.1189963
49. Mattison, S.M., Beheim, B., Chak, B., Buston, P. (2016) Offspring sex preferences among patrilineal and matrilineal Mosuo in Southwest China revealed by differences in parity progression. *Royal Society Open Science*. 3(9), 160526. DOI: 10.1098/rsos.160526
50. Stieglitz, J., Beheim, B., Trumble, B.C., Madiemenos, F.C., Kaplan, H. & Gurven, M. (2015) Low Mineral Density of a Weight-Bearing Bone Among Adult Women in a High Fertility Population. *American Journal of Physical Anthropology*. 156(4), pp.637-648. DOI: 10.1002/ajpa.22681
51. Blackwell, A.D., Tamayo, M.A., Beheim, B., Trumble, B.C., Stieglitz, J., Hooper, P.L., Martin, M., Kaplan, H., Gurven, M. (2015) Helminth infection, fecundity, and age of first pregnancy in women. *Science*. 350(6263) pp. 970-972. DOI: 10.1126/science.aac7902
52. Beheim, B., Thigpen, C. & McElreath, R. (2014) Strategic social learning and the population dynamics of human behavior: the game of Go. *Evolution and Human Behavior*. 35(5), pp. 351-357. DOI: 10.1016/j.evolhumbehav.2014.04.001
53. Beheim, B. & Baldini, R. (2012) Evolutionary decomposition and the mechanisms of cultural change. *Cliodynamics*. 3(2). DOI: 10.21237/C7clio3212123
54. Beheim, B. & Bell, A.V. (2011) Inheritance, ecology and the evolution of the canoes of East Oceania. *Proc. R. Soc. B*. 278(1721), pp. 3089-3095. DOI: 10.1098/rspb.2011.0060
55. Borgerhoff Mulder, M. & Beheim, B. (2011) Understanding the nature of wealth and its effects on human fitness. *Phil. Trans. R. Soc. B*. 366, pp.344-35. DOI: 10.1098/rstb.2010.0231

Research Group Leadership and Supervision

Postdoctoral supervision

- Dr. Peeter Tinitis, Postdoctoral Supervisor, June 2024–present, MPI-EVA.
- Dr. Oleg Sobchuk, Postdoctoral Supervisor, February 2023–February 2026, MPI-EVA.
- Dr. Daniel Redhead, Postdoctoral Supervisor, September 2018–August 2020, MPI-EVA.

Doctoral supervision and committees

- Dr. Riana Minocher, Doctorate Supervisor, January 2018–November 2022, University of Leipzig / MPI-EVA.

- Dr. N. Han Tran, Doctorate Supervisor, October 2018–February 2022, University of Leipzig / MPI-EVA.
- Dr. Natalia Fedorova, Doctoral Committee Member, August 2017–November 2024, University of Leipzig / MPI-EVA.
- Anja Becker, Dissertation Thesis Advisory Committee, August 2026–present, MPI-EVA.

Master's supervision

- Anja Becker, Master's Thesis Supervisor, completed November 2020, University of Leipzig.
- Leonie Ette, Master's Thesis Supervisor, completed April 2020, University of Leipzig.

Undergraduate supervision

- Anja Becker, Undergraduate Thesis Supervisor, completed September 2018, University of Leipzig.

Teaching Experience

- **Scientific Computing in Evolutionary Anthropology**, Primary Instructor, taught annually 2019–2026.
- **The Human Life Course** (ANTH 160), Primary Instructor, 2013, 2014, 2015.
- **Human Behavioral Ecology** (ANTH 360), Primary Instructor, 2014.
- **Research Computing in Evolutionary Anthropology** (ANTH 560), Primary Instructor, 2014.
- **Workshop on Statistical Rethinking**, instructor at University of New Mexico, 2016.
- **Evolution and Human Nature** (ANT 50), Primary Instructor, 2011.
- **Applied Research Methods** (ESP 178), Primary Instructor, 2011, 2012.
- **Cultural Ecology** (ANT 102), Primary Instructor, 2010.

Field Research

- **2024:** Tsimane' near-river communities, Bolivia (3 weeks); Mayangna and Miskito communities, Nicaragua (4 weeks).
- **2023:** Tsimane' near-river communities, Bolivia (2 weeks).
- **2019:** Tsimane' near-river communities, Bolivia (3 weeks).
- **2018:** Tsimane' near-river communities, Bolivia (6 weeks).
- **2017:** Tsimane' near-river communities, Bolivia (5 weeks); Hokkaido dairy communities, Japan (11 weeks).
- **2014:** Tsimane' near-river communities, Bolivia (2 months).
- **2013:** Tsimane' upriver communities, Bolivia (2 months).

Selected Invited Talks and Conference Presentations

- **2022.** "Strategic Social Learning in the Era of Superhuman Artificial Intelligence: A Large-Scale, Longitudinal Case Study using the Game of Go," Cultural Evolution Society Meeting.
- **2020.** "Improving Reproducibility in Evolutionary Anthropology," Santa Fe Institute ENDOW workshop.
- **2019.** "Open Science in the Field," Open Science and Reproducibility workshop, Interacting Minds Centre, Aarhus University.
- **2019.** "The Adaptive Logic of False Beliefs," Human Behavior and Evolution Society Meeting.
- **2018.** "Atherosclerosis as a Culturally-Transmitted Disease among Acculturating Horticulturalists," European Human Behavior and Evolution Association Meeting.
- **2017.** "Beyond conformity in cultural diffusion: Knowledge, opportunity and motivation," Cultural Evolution Society Meeting.
- **2017.** "Advances in studying cultural diffusions," Department of Behavioral Sciences, Hokkaido University.

- **2016.** “Transparency and reproducibility in observational research: lessons from anthropology,” Yale’s Day of Data, with Melanie Martin.
- **2016.** “Anthropological Research as Software Development,” Yale Department of Anthropology.
- **2013.** “Evolutionary dynamics in the Game of Go,” Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- **2012.** “Strategic Social Learning and Evolutionary Arms Races in the Game of Go,” invited talk, Santa Fe Institute.
- **2011.** “Developing Measures of Cultural Variation and Dynamics in Hokkaido, Japan,” Radcliffe Institute, Harvard University.

Academic Leadership and Service

- Program Committee, Cultural Evolution Society 2026 Rabat.
- Co-Investigator and Data Director, Tsimane Health and Life History Project, 2012–present.
- Data Curation Supervisor, Department of Human Behavior, Ecology and Culture, MPI-EVA, 2016–2025.
- MPI-EVA Worker’s Council (*Betriebsrat*) reserve member, elected November 2018–November 2019.
- Website Editor, Evolutionary Anthropology Society, 2016–2021.

Grant review

NSF Cultural Anthropology; CES ACE Award Program 2025

Peer review

Evolutionary Human Sciences; Royal Society Open Science; Current Anthropology; Journal of Theoretical Biology; Advances in Complex Systems; Human Nature; Science Advances; Cliodynamics; PLOS ONE; Evolution and Human Behavior.

Public Engagement and Open Science

- “Planck’s Principle, Price’s Theorem, and the Forces of Cultural Evolution,” UCLA BEC.
- “Human Innovation in the AI Age,” The Cognitions Podcast.
- “Cultural Dynamics with Bret Beheim,” Evolutionary Psychology (The Podcast).
- “Cycles, Cohorts, and the Zeitgeist,” Max Planck Institute for Human Development.
- Beheim, B. (2017) “Text Mining the Human Behavioral Ecology Bibliography.”
- Beheim, B. (2017) “Generating Ties to Explore Social Network Connectivity.”

Professional Memberships and Certification

- Software Carpentry / Data Carpentry certification, since 2018.
- Cultural Evolution Society, last renewed 2026.
- Human Behavior and Evolution Society, last renewed 2026.
- European Human Behavior and Evolution Association, last renewed 2026.
- American Association of Biological Anthropologists, last renewed 2026.

Languages and Quantitative Methods

- **Languages:** German, intermediate (B1 certificate); Japanese, intermediate; Spanish, intermediate; Tsimane’, basic.
- **Computing and quantitative tools:** R, Stan, Python, SQL, Stata, MATLAB, UNIX shell.